



Technical Note

Verification of the modified point kinetics model in the MARS code for molten salt reactors

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ABSTRACT

This study presents the implementation and verification of a modified point kinetics model (PKM) within the thermal-hydraulic system code MARS for application to liquid-fueled molten salt reactors (MSRs). Traditional PKMs for MSRs establish the delayed neutron precursor (DNP) balance equation by simplifying the reactor system into two regions: a reactor core and an external loop only. This approach is inadequate for realistic MSR systems with complex flow paths. To overcome this limitation, a three-dimensional DNP transport equation was introduced into the PKM instead of DNP balance equation, enabling a more accurate treatment of precursor distribution throughout the reactor system. The model was numerically solved in the MARS code using the upwind scheme, and the effective delayed neutron fraction was evaluated from steady-state precursor distributions. Verification of the modified PKM in the MARS code was performed through comparison with exact analytical solutions and the independently developed GAMMA + code. The results confirm the correctness and numerical accuracy of the modified PKM and demonstrate the MARS code's capability for both steady-state and transient analysis of liquid-fueled MSRs.

1. Introduction

Molten salt reactors (MSRs) were actively developed between the 1950s and 1970s, primarily at Oak Ridge National Laboratory. MSRs offer several operational and safety advantages over solid fueled designs [1,2]. They can operate at high temperatures with low-pressure molten salt coolant, enable continuous fuel recycling, and produce minimal long-lived nuclear waste. However, MSR development in the 1970s faced significant challenges, including material corrosion, immature fuel reprocessing technology, and economic competition from well-established light water reactors. In recent years, advances in materials science and fuel reprocessing have addressed the previous technical limitations, and the increasing demand for sustainable and efficient energy solutions has renewed global interest in MSR technology.

The thermal-hydraulic and neutronic characteristics of MSRs differ significantly from those of light water reactors. Notably, in a liquid-fueled MSR, the fuel and fission fragments circulate within the reactor system along with the molten salt. As a result, delayed neutron precursors (DNPs) also flow with the molten salt, which leads to a

significant impact on reactor kinetics due to the loss of delayed neutrons. Nuclear fission energy is released directly into the liquid coolant, acting as an internal heat source, and radioactive fission products (RFPs), which generate decay heat, are also transported throughout the reactor system. These unique characteristics must be carefully considered in both reactor kinetics and thermal-hydraulic models. Wu et al. [3] conducted a comprehensive and valuable literature review on modeling these coupled features.

Numerous safety analysis codes have been developed for MSRs. Some of these codes were specifically developed for MSRs [4–8], while others are modifications of existing thermal-hydraulic system codes designed for solid-fueled reactors [9–16]. Most of these codes employ a point kinetics model (PKM) [17] to provide a practical means of predicting and analyzing reactor power behavior under transient conditions. A PKM consists of a set of differential equations that relate the rate of change of neutron population to reactivity and DNPs in a nuclear reactor. DNPs are typically divided into six groups based on their half-lives. To incorporate the effects of liquid fuel flow in an MSR, the DNP balance equation requires modification. In earlier models [18–20], the reactor coolant system was simplified by dividing it into a reactor

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core and an external loop. Subsequently, two terms were added to the DNP balance equation to account for the effects of outflow from the core and inflow from the external loop. For convenience, this method is referred to as the “two-region approach.” In this approach, accurate solutions depend heavily on the fuel transit times in the core and the external loop. However, a real MSR system is more complex than just two regions; it often includes bypass-flow paths for chemical processing, helium bubbling, and other processes. Consequently, it can be challenging to accurately define the fuel transit time in the external loop. Furthermore, under certain transient conditions, such as during fuel drainage for safety, even the transit time in the core can be difficult to determine. Thus, the two-region approach has practical limitations when applied to real MSR systems. Nonetheless, this method, and some improved variations, remains in use in several studies due to its simplicity [4–15]. However, the problem can be effectively resolved by solving the DNP transport equation throughout the reactor system [16, 21].

The authors have modified the thermal-hydraulic system code MARS [22,23] for application to liquid-fueled MSRs [21]. This approach, which involves modifying the existing code, allows for the full utilization of MARS’s capabilities. The MARS code is based on the consolidated version of RELAP5/MOD3 and COBRA-TF codes [24–27]. It employs a two-fluid model for two-phase flows and incorporates various component models and special process models. MARS is known for its versatility, robustness, and systematic validation [22,23,28]. However, as it was initially developed for light water reactors, certain modifications are necessary for MSRs. Initially, the properties of molten salts, such as KCl-UCl₃, LiF-BeF₂-ZrF₄-UF₄, and FUNaK, were integrated into the code. Subsequently, to address the aforementioned limitations, a modified PKM with six groups of delayed neutrons was implemented, where a three-dimensional DNP transport equation is solved across the reactor system. Furthermore, the transport equation for eleven groups of RFPs was implemented to calculate the decay heat at each computational cell, which is then used as an internal heat source for the two-fluid energy equations.

This paper focuses on the implementation and verification of the modified PKM within the MARS code. Section 2 introduces the modified PKM, along with the three-dimensional DNP transport equation and the numerical solution method. Section 3 presents the verification results, including comparisons between MARS calculations and three exact solutions. The results of a code-to-code comparison are also provided. Finally, the conclusions are summarized in the last section.

2. Modified point kinetics model and numerical solution method

This section introduces the modified PKM for a liquid-fueled MSR. Subsequently, the numerical solution method and the steady-state solution are presented. The RFP transport equation is also briefly introduced.

2.1. Point kinetics model for a liquid fueled MSR

Similar to other thermal-hydraulic system codes, the MARS code employs a PKM designed for solid-fueled reactors, incorporating six groups of DNPs [22]. However, for liquid-fueled MSRs, the PKM must account for the flow effect of DNPs. In earlier models [18–20], the neutron and DNP population balance equations were formulated as follows:

$$\frac{dN(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} N(t) + \sum_{i=1}^6 \lambda_i C_i(t) \quad (1)$$

$$\frac{dC_i(t)}{dt} = \frac{\beta_i}{\Lambda} N(t) - \lambda_i C_i(t) - \frac{C_i(t)}{\tau_c} + \frac{C_i(t - \tau_L)}{\tau_c} e^{-\lambda_i \tau_L} \quad (2)$$

where β_{eff} is the effective delayed neutron fraction, which is discussed in Sec. 2.3. τ_c and τ_L are the fuel transit times in the core and the external loop, respectively. The third and fourth terms on the right-hand side of Eq. (2) are added to account for the flow effect. However, this approach is only valid when the reactor system consists solely of the core and the external loop. In a real MSR system, there may be other bypass-flow paths, such as those for chemical processing and helium bubbling. In such cases, accurately defining the fuel transit time in the external loop becomes challenging. Considering the short half-lives of DNPs, even a minor error in τ_L can lead to significant inaccuracies in reactor kinetics calculations. Under certain transient conditions, such as during fuel drainage operations, even τ_c can be difficult to determine. This also evokes inaccuracies.

This problem can be readily resolved by solving a three-dimensional DNP transport equation throughout the reactor system including the reactor core and all the fluid systems directly connected to it, instead of using Eq. (2):

$$\frac{\partial c_i}{\partial t} + \nabla \cdot (c_i \nu_f) = f \frac{\beta_i}{\Lambda} n(t) - \lambda_i c_i \quad (3)$$

where c_i and $n(t)$ are the densities of DNPs and neutrons, respectively. β_i in Eq. (3) is the same as β_i in Eq. (2). The coefficient f represents the power distribution factor within the reactor core and becomes zero outside of the reactor core. This formulation relies on the assumptions that the properties of the molten salt are not affected by the presence of the DNPs, and that the DNPs are transported solely by the molten salt at its velocity. The diffusion term was excluded because its effect is negligibly small [16,29]. Conceptually, Eq. (3) is analogous to the boron transport equation [26] in the MARS and RELAP5 codes, with the exceptions of the generation and decay terms on the right-hand side.

After numerically solving Eq. (3) for the entire reactor system, the number of DNPs within the reactor core, denoted as $C_i(t)$, can be determined using:

$$C_i(t) = \int_{Core} c_i(t) dV \text{ or } C_i(t) = \sum_{k=1}^{NC} c_{i,k}(t) V_k \quad (4)$$

where NC is the number of thermal-hydraulic meshes for the reactor core. Consequently, Eq. (2) is replaced by Eqs. (3) and (4) in the modified PKM.

The transient reactivity $\rho(t)$ in Eq. (1) incorporates the effects of control rods, poisons, and the feedback from fuel temperature, fuel density, and reflector temperature:

$$\rho(t) = \rho_{rod} + \rho_{poison} + \alpha_{FT}(T_F - T_{F0}) + \alpha_{FD}(\rho_F - \rho_{F0}) + \alpha_{RT}(T_R - T_{R0}) \quad (5)$$

where α is a reactivity coefficient.

2.2. Numerical solution method

The DNP transport equation can be solved using either the upwind difference scheme or a second-order accurate Godunov scheme, as implemented in the MARS and RELAP5/MOD3 codes. However, for the sake of simplicity and robustness in handling complex transient flow scenarios and/or three-dimensional modeling of the reactor core’s thermal hydraulics, the upwind difference scheme was chosen. Applying this scheme [22,26] to Eq. (3) for a computational volume k yields:

$$V_k \frac{c_{i,k}^{n+1} - c_{i,k}^n}{\Delta t} + \sum_{j(k)=1}^{NO} \left(c_{i,j(k)}^n v_{f,j(k)}^{n+1} A_{j(k)} \right) - \sum_{j=NO+1}^{NO+NI} \left(c_{i,j(k)}^n v_{f,j(k)}^{n+1} A_{j(k)} \right) = V_k \left(f_k \frac{\beta_i}{\Lambda} n^n - \lambda_i c_{i,k}^n \right) \quad (6)$$

where NO and NI are the numbers of the outlet and inlet junctions connected to the volume k , respectively. f_k is the power distribution

factor of volume k . $\dot{c}_{ij}$ is a donor quantity at junction j . The generalized convective term in Eq. (6) allows the use of three-dimensional meshes in modelling the reactor core's thermal-hydraulics, which is essential for subsequent coupling with a three-dimensional reactor kinetics code.

Eq. (6) is coupled with the fluid flow through the liquid-phase velocity $v_{fj(k)}^{n+1}$. During each time-step advancement, the governing equations for fluid flow are solved first in the MARS code. Subsequently, Eq. (6) is solved to obtain $c_{i,k}^{n+1}$ using the new liquid-phase velocity $v_{fj(k)}^{n+1}$. This sequence mirrors that used for solving the boron transport equation in the MARS code. The number of the i -th group DNP in the reactor core, $C_i(t)$, is calculated using Eq. (4). It is then incorporated into the neutron balance equation, Eq. (1). To solve Eq. (1), the prompt jump approximation is applied as follows:

$$N(t) \approx \frac{\Lambda}{\beta_{\text{eff}} - \rho(t)} \sum_{i=1}^6 \lambda_i C_i(t) \quad (7)$$

2.3. Initial steady-state condition and the effective delayed neutron fraction

To obtain an initial steady state for an MSR system, a constant core power is given and null transient advancements for the fluid flow and the DNP transport, are carried out. This means that Eq. (1) is not solved for a steady-state calculation. When the flows in the entire thermal-hydraulic system reach a steady state, the distributions of the DNPs in the system are also fixed. The steady-state DNPs in the reactor core, C_{io} , can be determined using Eq. (4). Since dN/dt and $\rho(t)$ in Eq. (1) become zero in a steady state, β_{eff} can be obtained as:

$$\beta_{\text{eff}} = \frac{\Lambda}{N_o} \sum_{i=1}^6 \lambda_i C_{io} \quad (8)$$

The effective delayed neutron fraction β_{eff} in Eq. (8) is applied in subsequent transient calculations.

It is noted that β_{eff} in Eq. (8) is slightly smaller than the exact solution of β_{eff} [9] from the system of point kinetics equations, i.e., Eqs. (1) and (2), which is given as:

$$\beta_{\text{eff}} = \sum_{i=1}^6 \frac{1}{1 + \frac{1}{\lambda_i \tau_c} (1 - e^{-\lambda_i \tau_c})} \beta_i \quad (9)$$

In Eq. (2), the DNPs transit time in the core is assumed to be τ_c . However, for example, if the DNPs are generated in the upper half of the reactor core, their transit times in the core are much shorter than τ_c , leading to a greater loss of the delayed neutrons. Eq. (9) cannot consider this situation and, therefore, tends to overestimate the effective delayed neutron fraction. Meanwhile, Eq. (8) realistically considers the spatial distribution of DNPs in the reactor core.

2.4. Decay heat model

Because the RFPs flow through the system, they are not confined to the core and the decay heat is released outside the core. To model this, a transport equation for RFPs can be formulated similarly to the DNP transport equation in Eq. (3):

$$\frac{\partial h_i}{\partial t} + \nabla \cdot (h_i \nu_f) = f \gamma_i n(t) - \lambda_i h_i \quad (10)$$

where h_i and γ_i are the i -th group RFP density and yield, respectively. Numerical solutions for Eq. (10) are obtained at the end of a time-step advancement for the fluid flow using the same method as employed for the DNP transport equation.

The sum of fission power and decay heat in a computational cell k can be expressed as:

$$Q_k(t) = \kappa_e \left(f_k N(t) + \sum_i \lambda_i h_{i,k} V_k \right) \quad (11)$$

where κ_e is an energy conversion factor. The quantity $Q_k(t)$ is transferred to the liquid-phase energy equation as an internal heat source at each time step.

3. Verification of the modified point kinetics model

Verification is the process of assessing the correctness and numerical accuracy of the solution to a given physical model defined by a set of equations [30,31]. In other words, verification evaluates whether the equations are solved correctly by the code. Validation, on the other hand, is the process of assessing the code's physical accuracy by comparing simulation results with experimental data.

Section 3 presents the verification of the modified PKM implemented in the MARS code. Three exact solutions for a conceptual problem are derived and compared with the results from MARS calculations. Additionally, a code-to-code comparison with the GAMMA + code [32] is performed.

For all calculations in Section 3, properties of molten salt KCl-UCl₃ were used [21,32]. The delayed neutron parameters employed for verification are listed in Table 1, and the prompt neutron lifetime Λ was assumed to be 10^{-4} s.

3.1. Steady-state DNP distributions

Fig. 1 depicts a conceptual, vertical pipe reactor with a diameter of 0.5 m, a reactor core height of 1 m, and a riser of 0.2 m. The steady-state core power is 5 MW. Fresh molten salt fuel is introduced to the inlet at the bottom ($z=0$ m) at a constant velocity of 1 m/s. Consequently, the DNP densities at the inlet are zero, i.e., $c_i(0) = 0$. A constant pressure is maintained at the top of the reactor. In this reactor system, τ_c is 1 s and τ_L is infinite.

Under the assumptions of a uniform power distribution within the core and a uniform flow velocity, Eq. (3) can be expressed as follows:

$$v_f \frac{dc_i(z)}{dz} = f \frac{\beta_i}{\Lambda} n_o - \lambda_i c_i(z) \quad (12)$$

where f is 1 in the core and zero in the riser. The exact solution of Eq. (12) can be obtained as follows:

$$c_i(z) = \begin{cases} \frac{\beta_i}{\lambda_i \Lambda} n_o (1 - e^{-(\lambda_i/v_f)z}) & \text{for the core, } z \leq z_c, \\ c_i(z_c) e^{-(\lambda_i/v_f)z} & \text{for the riser, } z > z_c. \end{cases} \quad (13)$$

In the MARS calculations, uniform axial meshes of 12, 30, and 60 divisions were used for the core and riser regions. The results of the six-group DNP distributions obtained from MARS are compared with the exact solutions in Fig. 2. The comparison shows good agreement between the MARS results and the exact solutions. Fig. 3 provides a partial enlargement of Fig. 2, illustrating that the accuracy improves when finer meshes are used. A similar approach can be employed to verify the RFP transport equation.

Table 1
The delayed neutron parameters employed for verification.

Group	β_i	λ_i (s ⁻¹)	half-life (s)
1	1.69E-04	3.87E+00	0.179
2	8.32E-04	1.40E+00	0.495
3	2.64E-03	3.11E-01	2.229
4	1.22E-03	1.15E-01	6.027
5	1.38E-03	3.17E-02	21.866
6	2.47E-04	1.27E-02	54.579

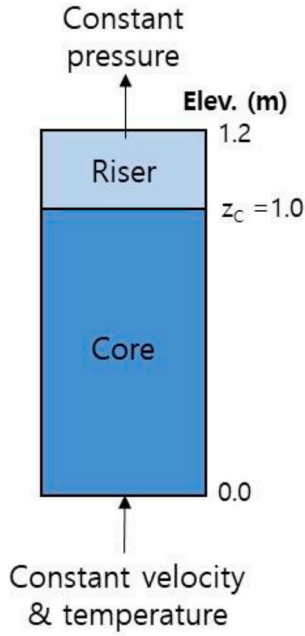


Fig. 1. Schematic of a conceptual vertical pipe reactor.

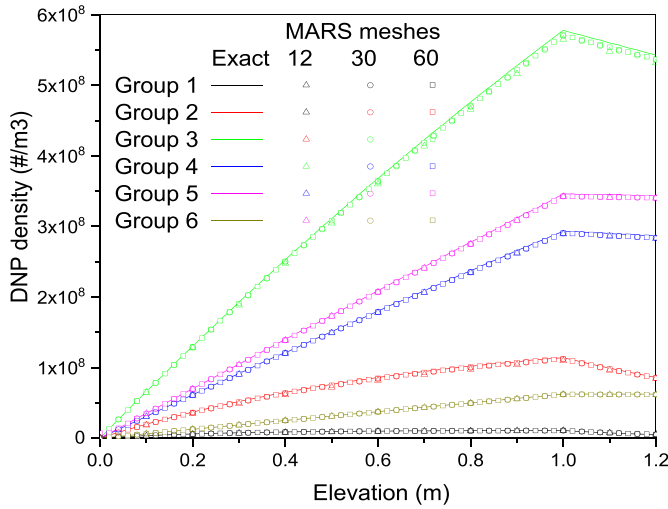


Fig. 2. Comparison of the DNP distributions: the MARS and exact solutions.

3.2. Effective delayed neutron fraction

The effective delayed neutron fraction of the conceptual reactor in Section 3.1 can be determined analytically. By integrating Eq. (13) over the reactor core, the steady state number of i -th group DNP in the reactor core, C_{io} , is derived as follows:

$$C_{io} = \int_0^{z_c} c_i(z) A_C dz$$

$$= \frac{\beta_i}{\lambda_i \Lambda} n_o A_C \left(z_c - \frac{1}{\lambda_i / v_f} (1 - e^{-(\lambda_i / v_f) z_c}) \right) \quad (14)$$

Inserting Eq. (14) into Eq. (8) yields an exact solution of β_{eff} :

$$\beta_{eff} = \frac{\Lambda}{N_o} \sum_{i=1}^6 \lambda_i C_{io}$$

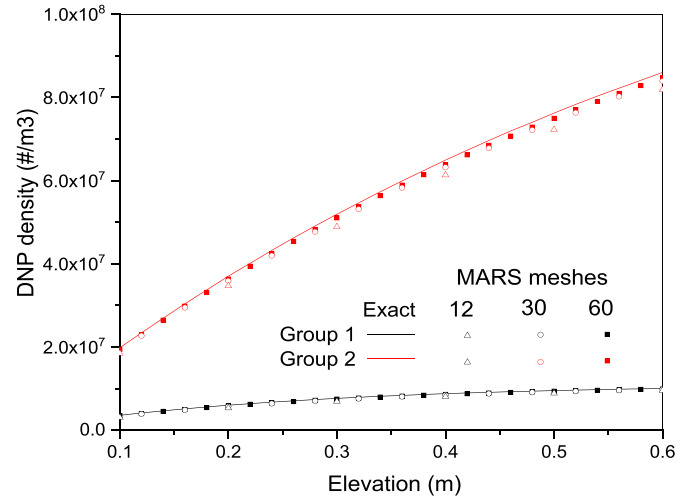


Fig. 3. Comparison of the DNP distributions: the MARS and exact solutions (Partially enlarged).

$$= \sum_{i=1}^6 \beta_i \left(1 - \frac{1}{(\lambda_i / v_f) z_c} (1 - e^{-(\lambda_i / v_f) z_c}) \right) \quad (15)$$

Table 2 compares the effective delayed neutron fraction obtained from MARS calculations with the exact solution in Eq. (15). For a case with 10 axial meshes for the core, the error exceeds expectations, primarily due to relatively large inaccuracies in the MARS calculation of short-lived DNPs, such as group 1 and 2 (see Fig. 3). However, when finer meshes are used, the error decreases progressively.

3.3. A transient solution

The conceptual reactor described in Section 3.1 is used again to derive an exact solution for a hypothetical transient. If the reactor core is modeled using a single control volume, Eq. (3) for the reactor core can be expressed as follows:

$$V_C \frac{dc_i(t)}{dt} = \left(\frac{\beta_i}{\Lambda} n(t) - \lambda_i c_i(t) \right) V_C - c_i(t) v_f A_C \quad (16)$$

where V_C and A_C are the volume and cross-sectional area of the reactor core, respectively. The steady-state solution of Eq. (16) is given by:

$$c_{io} = \frac{\beta_i}{(\lambda_i + v_f / z_c) \Lambda} n_o \quad (17)$$

If the nuclear fission instantaneously ceases at $t = 0$ s, Eq. (16) simplifies to:

$$\frac{dc_i(t)}{dt} = - \left(\lambda_i + \frac{v_f}{z_c} \right) c_i(t) \quad (18)$$

The exact solution of Eq. (18) is given by:

Table 2
Comparison of the effective delayed neutron fractions.

Case	Exact solution from Eq. (15)	MARS calc.: # of meshes for the core (mesh length, cm)			
		10 (10 cm)	25 (4 cm)	50 (2 cm)	100 (1 cm)
β_{eff}	0.000972	0.001222	0.001119	0.001078	0.001056

Note: The exact β_{eff} from Eq. (9) with $\tau_C = 1$ s and $\tau_L = \infty$ is 0.001417 (overestimated in comparison to the above values).

$$c_i(t) = c_{i0} e^{-(\lambda_i + v_f/z_c)t} \quad (19)$$

This hypothetical transient was simulated using the MARS code with a single mesh for the reactor core. Fig. 4 shows that the DNP densities obtained from MARS are in good agreement with the exact solutions.

The results in Sections 3.1 to 3.3 demonstrate that the DNP transport equation is correctly solved in the MARS code.

3.4. Comparison with the GAMMA + code

For the purpose of code-to-code comparison, a simplified MSR design was first developed. Steady-state and transient analyses of the MSR system were then carried out using both the MARS and GAMMA + codes. Originally developed for the thermal-hydraulic analysis of gas-cooled reactors, the GAMMA + code [32] has since been adapted for application to MSRs. It incorporates a drift-flux model for two-phase flow and a modified point kinetics model that includes the transport equation for DNPs. Both MARS and GAMMA + employ a semi-implicit finite-difference method, using a staggered-grid mesh and a donor-cell scheme. The GAMMA + code has undergone extensive verification and validation for MSR applications [33–35].

3.4.1. A simple MSR design for code verification

The simple MSR consists of a 10 MW reactor core, a primary cooling system (PCS), and a secondary cooling system. The PCS includes a heat exchanger, a centrifugal pump, an expansion tank filled with helium, and associated piping. Heat generated in the reactor core and PCS is transferred to the secondary side via the heat exchanger. On the secondary side, a constant flow of molten salt is maintained to cool the PCS. The working fluid for both the primary and secondary sides is KCl-UCl₃ molten salt. Basic data for the reactor design are presented in Ref. [21].

Fig. 5 shows the MARS nodalization of the system. Component 100 represents the reactor core, which consists of 7 axial meshes. Component 580 is a connecting pipe between the riser and the expansion tank. The primary side of the heat exchanger is represented by component 300. Junction 175 and drain valve 515 were created to test various flow paths and were not used in the calculations for verification.

3.4.2. The results of a steady-state calculation

A steady state was achieved through a null transient calculation of 2000 s with fixed boundary conditions. Both the MARS and GAMMA + codes used the same geometry and boundary conditions. Table 3 presents the steady-state results from both codes. It was observed that the pump power was slightly higher in the GAMMA + code, while the

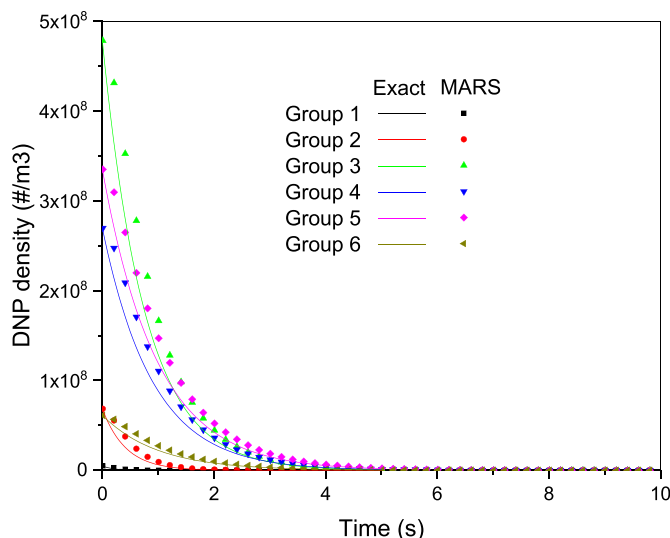


Fig. 4. Comparison of the DNP densities during a hypothetical transient.

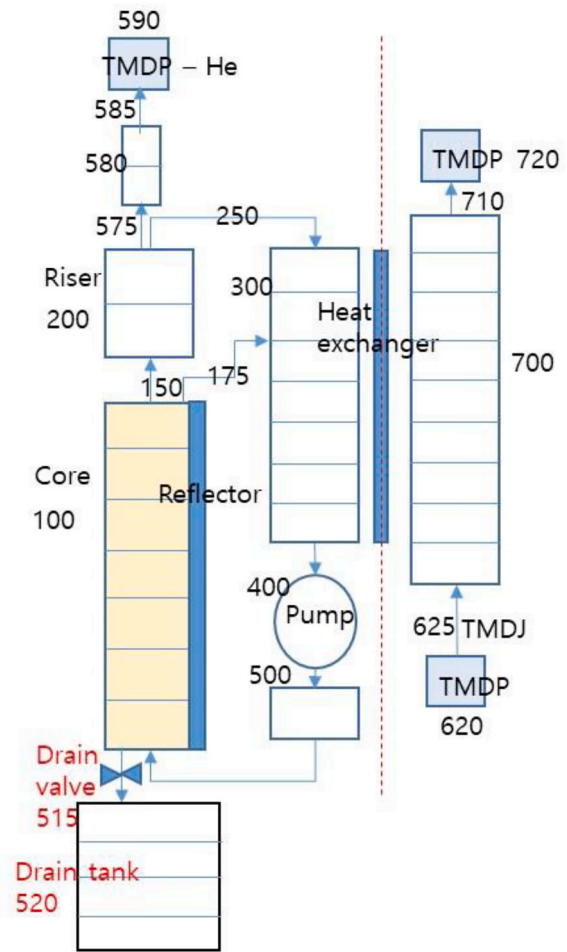


Fig. 5. The MARS nodalization for a conceptual MSR design.

Table 3

Steady-state calculation results.

Parameter	MARS	GAMMA+
Power (MWt)	10	10
Core inlet temp. (K)	868.56	868.28
Core exit temp. (K)	907.95	908.14
Primary mass flow rate (kg/s)	514.31	514.00
β_{eff} (–)	0.003266	0.003263
Secondary side inlet temp. (K)	815.50	815.50
Secondary-side exit temp. (K)	848.64	849.05
Secondary mass flow rate (kg/s)	644.33	644.33

thermal resistance of the heat exchanger was a little greater in the MARS code. Overall, the results from the two codes showed good agreement, with only minor differences in the core mass flow rate and, consequently, in β_{eff} .

Table 4 lists the steady-state 6-group DNP distributions across the seven axial meshes of the reactor core. The maximum relative error between the two codes is 0.25 %. This discrepancy is primarily due to a small difference in the fluid velocity. The RFP distributions were also predicted with similar accuracy.

In the steady state, the flow in the connecting pipe (component 580) is stagnant and vertically stratified with the upper half filled with helium. If the RFPs remain in stagnant molten salt, continuous heating due to decay heat occurs. To avoid unrealistic temperature rise, the RFPs in component 580 were removed and its temperature was set equal to that of the riser (component 200) for subsequent transient analysis.

Table 4Steady-state DNP densities for 6 groups at the 7 axial meshes ($\#/\text{m}^3$).

(a) The MARS results						
M \ G	1	2	3	4	5	6
1	5.770E+06	4.890E+07	4.438E+08	6.844E+08	3.243E+09	1.501E+09
2	7.772E+06	7.773E+07	6.102E+08	7.607E+08	3.323E+09	1.514E+09
3	8.468E+06	9.487E+07	7.544E+08	8.327E+08	3.401E+09	1.526E+09
4	8.710E+06	1.051E+08	8.794E+08	9.008E+08	3.478E+09	1.539E+09
5	8.794E+06	1.112E+08	9.878E+08	9.651E+08	3.553E+09	1.551E+09
6	8.824E+06	1.148E+08	1.082E+09	1.026E+09	3.627E+09	1.564E+09
7	8.835E+06	1.169E+08	1.163E+09	1.083E+09	3.700E+09	1.576E+09
(b) The GAMMA+ results						
M \ G	1	2	3	4	5	6
1	5.772E+06	4.895E+07	4.438E+08	6.844E+08	3.244E+09	1.502E+09
2	7.770E+06	7.776E+07	6.102E+08	7.605E+08	3.322E+09	1.514E+09
3	8.463E+06	9.486E+07	7.543E+08	8.324E+08	3.399E+09	1.526E+09
4	8.703E+06	1.050E+08	8.790E+08	9.001E+08	3.475E+09	1.538E+09
5	8.787E+06	1.111E+08	9.871E+08	9.641E+08	3.549E+09	1.549E+09
6	8.816E+06	1.147E+08	1.081E+09	1.025E+09	3.621E+09	1.561E+09
7	8.827E+06	1.168E+08	1.162E+09	1.082E+09	3.692E+09	1.572E+09

3.4.3. Transient calculation results

A transient is initiated by two independent events; (i) reactivity changes from 100 s to 300 s as shown in Fig. 6 and (ii) pump trip at 600 s. All other boundary conditions remain unchanged. The reactivity coefficients in Eq. (5) for fuel temperature, fuel density, and reflector temperature were set to -1.160×10^{-5} (dk/k)/K, 4.110×10^{-5} (dk/k)/(kg/m³), and 4.080×10^{-5} (dk/k)/K, respectively. The effects of poisons were not considered.

The reactor system is subjected only to the reactivity changes until 600 s. Figs. 7 and 8 show the behaviors of the total power and molten salt temperature, respectively. These two parameters are strongly coupled via reactivity feedback. Both codes predict very similar results prior to the pump trip. However, since the thermal resistance was slightly higher in the MARS calculation, the molten salt temperature increased a bit more and later during the period of positive reactivity insertion between 100 s and 200 s. This indicates that the dynamic responses in the two codes are slightly different. Similar discrepancies are observed in the opposing case between 200 s and 300 s. These differences lead to small deviations in the core power.

At 600 s, the pump coast-down begins. Fig. 7 through 9 show that, after the pump trip, both codes predict very similar flow rates but exhibit slight differences in core power and temperature behaviors. These discrepancies are likely due to different dynamic responses of the heat exchanger in the two codes.

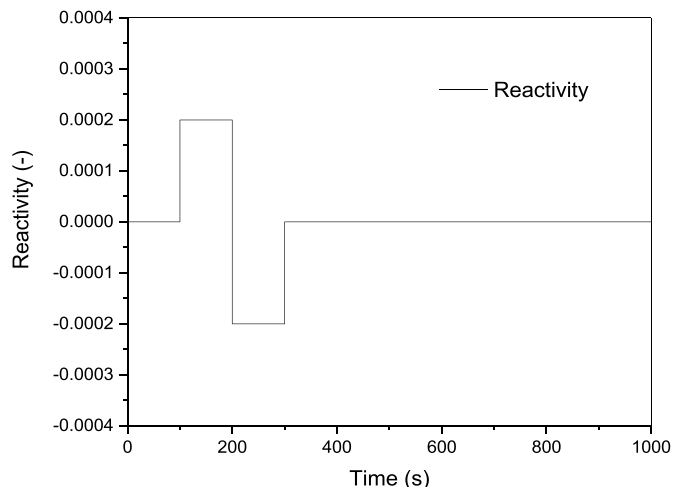


Fig. 6. The reactivity versus time.

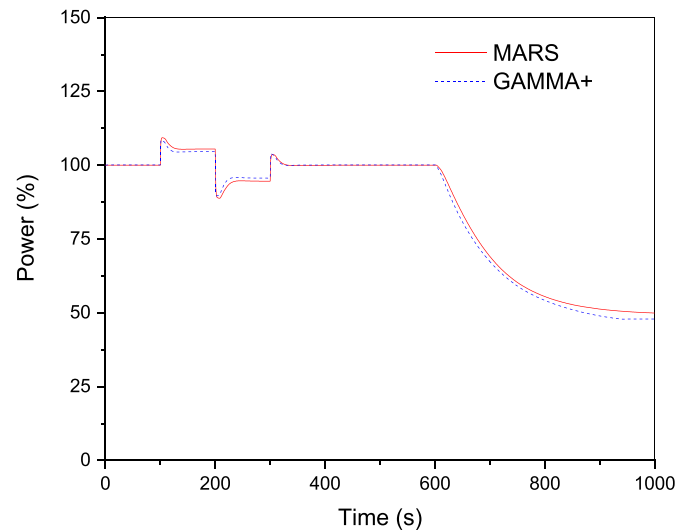


Fig. 7. Comparison of the total power behaviors.

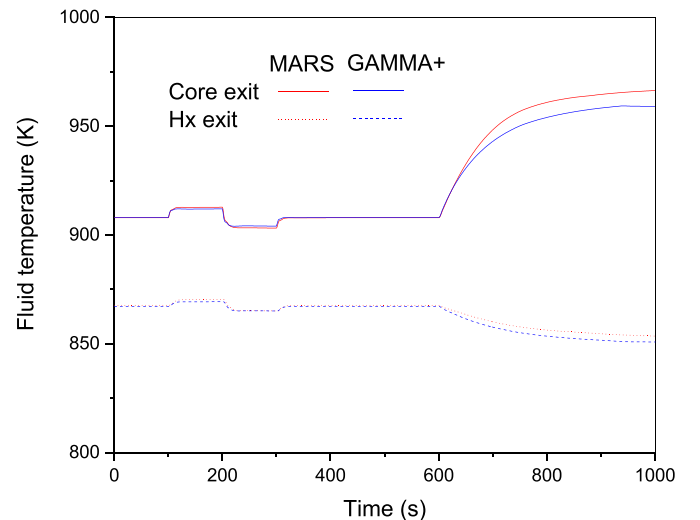


Fig. 8. The molten salt temperature behaviors.

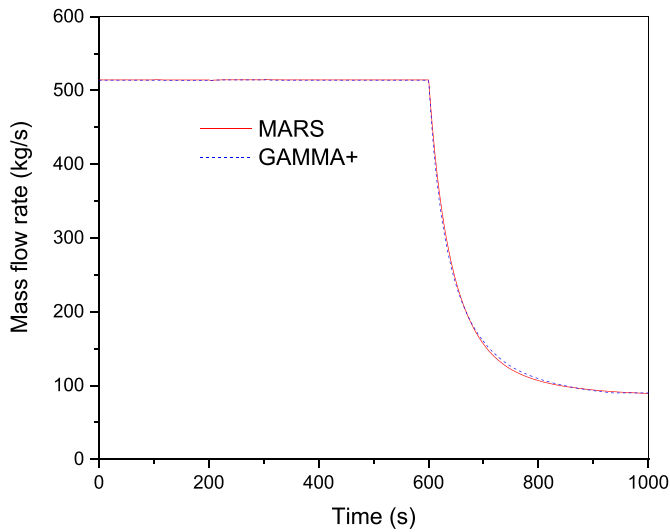


Fig. 9. Comparison of the core mass flow rates.

It is noteworthy that the two codes have been developed independently; nonetheless, they predict very similar results. Given that the GAMMA + code has undergone systematic verification and validation, this comparison indirectly verifies the MARS modifications for MSRs.

4. Concluding remarks

To extend the applicability of the MARS code to liquid fueled MSRs, property models for molten salts were implemented. Subsequently, a modified point kinetics model was developed to account for the effects of fuel flow specific to MSRs. Traditional point kinetics models for MSRs typically simplify the reactor system into two regions, a reactor core and an external loop, to establish the delayed neutron precursor balance equation. However, this simplification is inadequate for realistic MSR systems with complex flow paths. To address this limitation, a three-dimensional delayed neutron precursor transport equation was introduced into the modified point kinetics model. Additionally, a transport equation for radioactive fission products was incorporated to calculate local decay heat in an MSR system.

To verify the MARS code modifications, comparisons with exact solutions and a code-to-code comparison were conducted as follows:

- First, a conceptual MSR design was assumed. Two steady-state solutions and a transient solution for the delayed neutron precursors in the reactor were analytically derived. These exact solutions were compared with the results from MARS calculations to demonstrate the correctness and numerical accuracy of the code modifications. The results were satisfactory. It is also noted that these exact solutions can serve as valuable benchmarks for verifying any liquid-fueled MSR code.
- Next, a simplified MSR system design with basic components was devised. Steady-state and two independent transient scenarios, one involving reactivity changes and the other a flow transient due to the pump trip, were simulated using both the MARS and GAMMA + codes. The results of the code-to-code comparisons were very favorable for both the steady-state and transient analyses. Given that the two codes have been developed independently and yet produce very similar results, it can be concluded that the MARS modifications, including the modified point kinetics model and reactivity feedback mechanisms, are functioning correctly.

Currently, to establish the MARS code as a reliable tool for the design and safety analysis of MSRs, we are planning to undertake systematic validation as well as the development of MSR-specific thermal-hydraulic

models.

CRedit authorship contribution statement

Jae Jun Jeong: Writing – original draft, Validation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hong Sik Lim:** Validation. **Kwi Seok Ha:** Software, Methodology. **Yoon Jae Cho:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

A	Area (m^2)
c_i	Density of the i -th group DNP ($\#/\text{m}^3$)
C_i	Number of the i -th group DNPs in the reactor core (#)
f	Power distribution factor (–)
h	RFP density ($\#/\text{m}^3$)
k	Multiplication factor (–)
n	Neutron density ($\#/\text{m}^3$)
N	Number of neutrons in the reactor core (#)
t	Time (s)
v_f	Liquid-phase velocity (m/s)
V	Volume (m^3)
z	Spatial coordinate (m)
α	Reactivity coefficient (–)
β_i	Delayed neutron fraction of the i -th group DNP (–)
β_{eff}	Effective delayed neutron fraction (–)
γ	Radioactive fission product yield (–)
λ	Decay constant ($1/\text{s}$)
Λ	Prompt neutron life time (s)
ρ	Reactivity (–) or density (kg/m^3)
τ	Transit time (s)

Subscripts

C	Core
F	Fuel
FD	Fuel density
FT	Fuel temperature
i	i -th group
j	Junction index
k	Volume index
L	External loop
o	Steady state
R	Reflector
RT	Reflector temperature

Superscripts

n	Old time step
$n+1$	New time step

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